



Linear Algebra

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About me

Research and Services

Course and Office Hour Schedule

May need to adjust two instances of course

Policies

Teaching is fully English, QA can be Chinese
How to reach me
Course policy

Language

Mathematical Language A set of properties Do not remember what it is. Remember why we have them, and what we can use with them

Our world as a system: Game Movement

GTA

Our world as a system: Pipeline Building

Endfield

Our world as a system: Computer Graphics

AI model: parallel and sequential

Linear Algebra build what we are today

Ultimate Goal of Linear Algebra

We want to solve a system of equations:

$$\begin{cases} x_1 + 2x_2 = 5 \\ 2x_1 + 3x_2 = 8 \end{cases}$$

where:

- ▶ **Equation:** mathematical statement that asserts the equality of two expressions
- ▶ **System:** a collection of one or more equations involving the same set of variables
- ▶ **Solution:** find all values of the unknowns that satisfy every equation in the system

Example

Consider the system with 2 equations and 2 unknowns:

$$\begin{cases} 2x - y = 0 \\ -x + 2y = 3 \end{cases}$$

This system has the matrix form:

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

The coefficient matrix is A , the unknown vector is $\mathbf{x}$, and the right-hand side is $\mathbf{b}$.

Example cont.

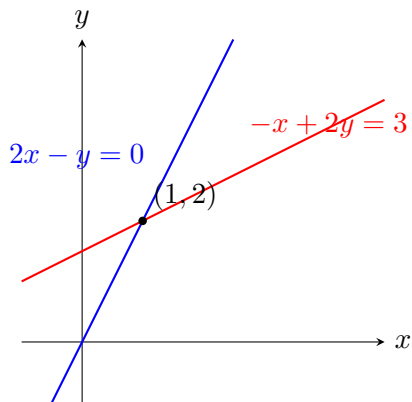
Keep the same equations, but look at them in three different ways:

- ▶ **Row picture:** each equation is a line in the plane
- ▶ **Column picture:** the solution is a combination of column vectors
- ▶ **Matrix picture:** write everything in compact linear algebra form

These viewpoints are equivalent, but each gives a different intuition.

Row picture

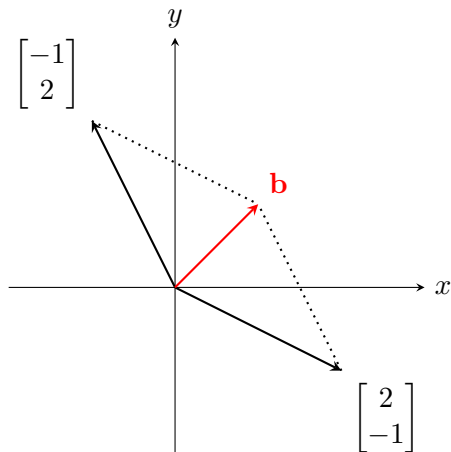
- ▶ $2x - y = 0$
- ▶ $-x + 2y = 3$



Column picture

The same system can be written as
a linear combination of vectors:

$$1 \begin{bmatrix} 2 \\ -1 \end{bmatrix} + 1 \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

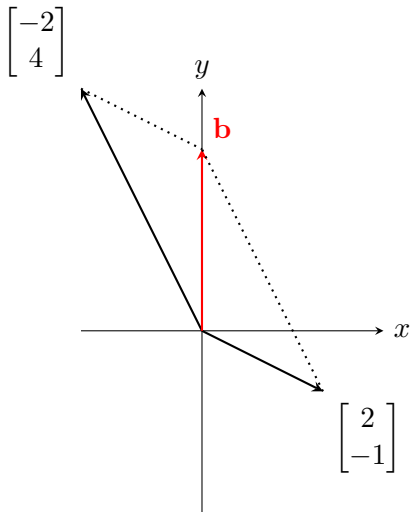


Column picture

$$x \begin{bmatrix} 2 \\ -1 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2x - y \\ -x + 2y \end{bmatrix}$$

$$x = 0.5$$

$$y = 1$$

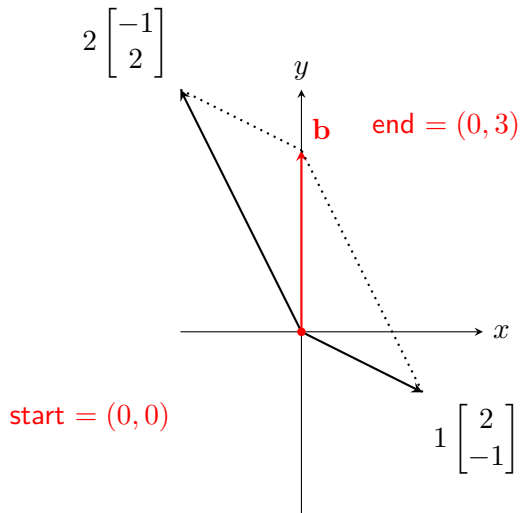


Column picture

Now: what is the
combination of x and y ?

$$x = 1$$

$$y = 2$$



Matrix

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$$\begin{array}{cc|c} 2 & -1 & 0 \\ -1 & 2 & 3 \end{array}$$

$$\begin{array}{cc|c} 2 & -1 & 0 \\ 0 & 3 & 6 \end{array}$$

Why do they work?

Linear

Okay, define linear for me.

Book Ours

Tricky. How to I tell?

A function is linear if it satisfies two properties:

- ▶ Additivity: $f(x + y) = f(x) + f(y)$
- ▶ Homogeneity: $f(cx) = cf(x)$

In the form of matrix...

Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 5 & -1 \\ 0 & 2 \end{pmatrix}.$$

Preserving Addition:

$$A + B = \begin{pmatrix} 1+5 & 2+(-1) \\ 3+0 & 4+2 \end{pmatrix} = \begin{pmatrix} 6 & 1 \\ 3 & 6 \end{pmatrix}.$$

Preserving Multiplication:

$$3A = 3 \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 3 & 6 \\ 9 & 12 \end{pmatrix}.$$

How many solutions can there be?

one unlimited none

Three examples

Existence of Solutions

Consistent vs Equivalent

"Good" matrix

$M \times N$ $N \times N$

Pre:Order of Matrix

"Enough" equations and "Just Enough" equations Simplified form (reduce equivalent)

Elementary Row Operations

Interchange Scaling Replacement (Addition)

Practice: into echelon form

What does echelon form look like

Terminologies

nonzero pivot pivot column pivot row "I am a pivot student" —¿ I hope you are all nonzero students...

Elementary Row Reduction

Leading variable and free variable

Existence and Uniqueness

Gaussian Elimination

